

UNIVERSITATEA TEHNICĂ A MOLDOVEI
FACULTATEA CALCULATOARE, INFORMATICĂ ȘI MICROELECTRONICĂ

Disciplina: Tehnici de programare
Lucrarea practică nr. 4.

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Grupa: SI-264;
Verificat: Reițman Patricia.

Chişinău, 2026

Capitolul 1 CODEDEX. Hello Word

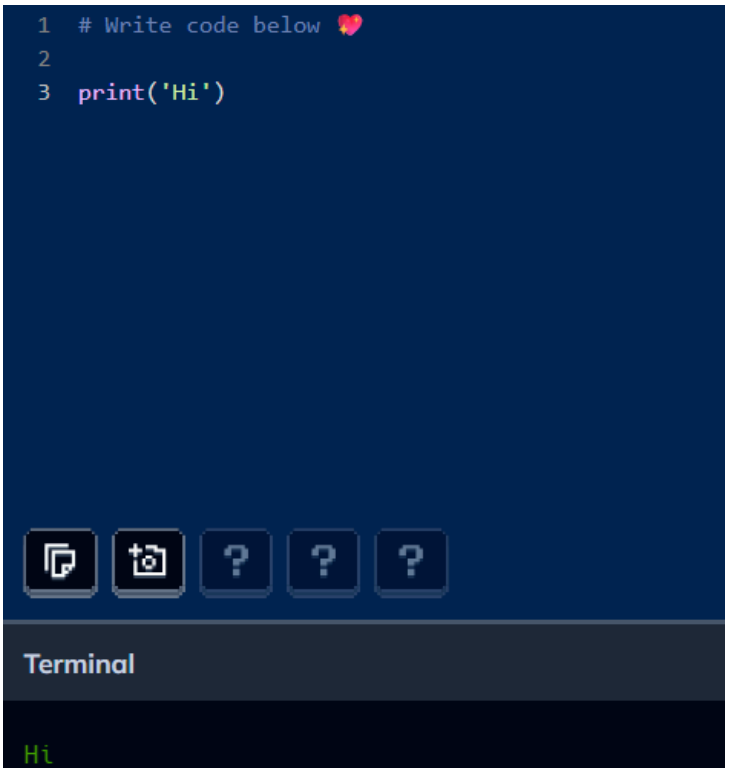


fig. 1.1 - Setting up

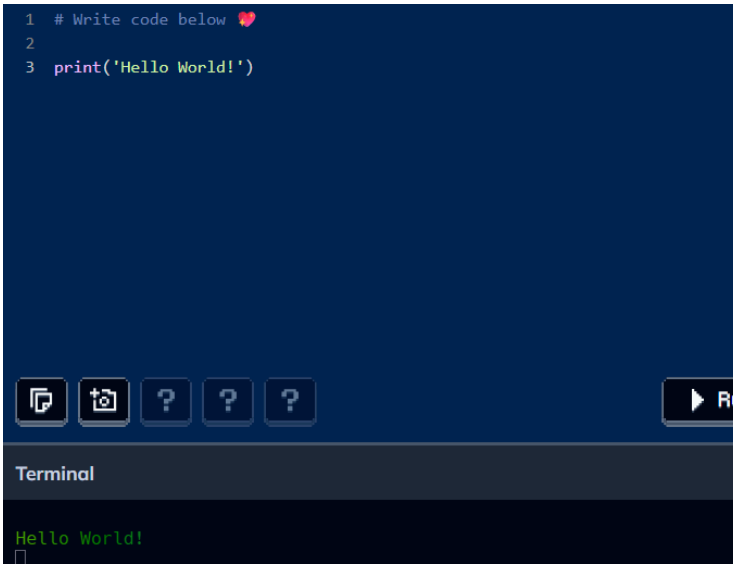


fig. 1.2 - Hello Word

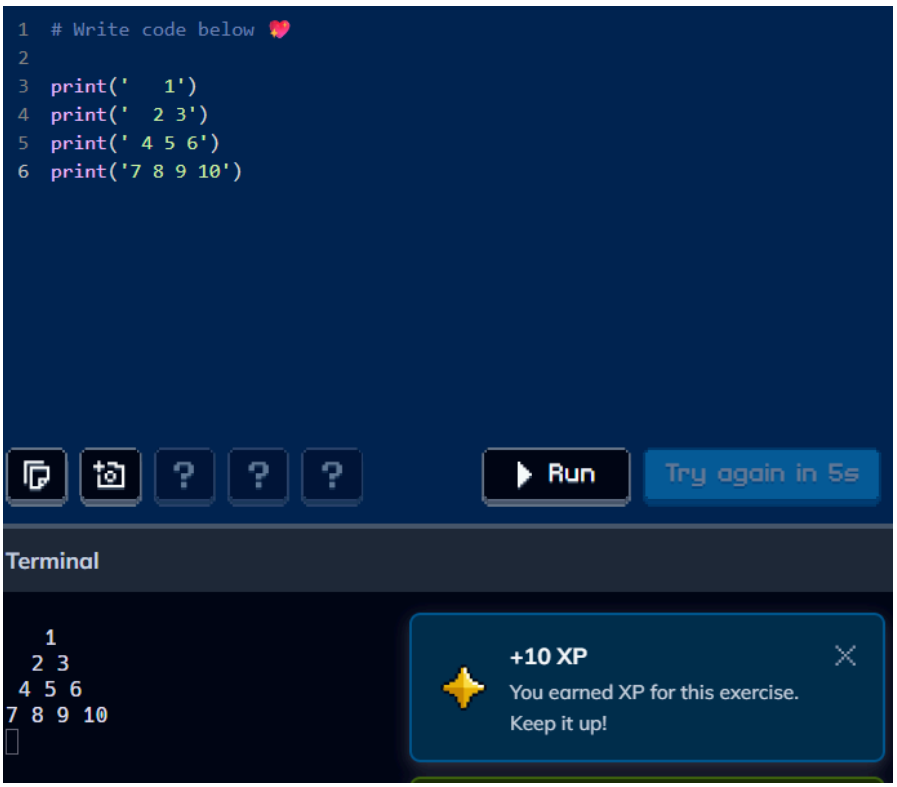


fig. 1.3 - Pattern



fig. 1.4 - Initials

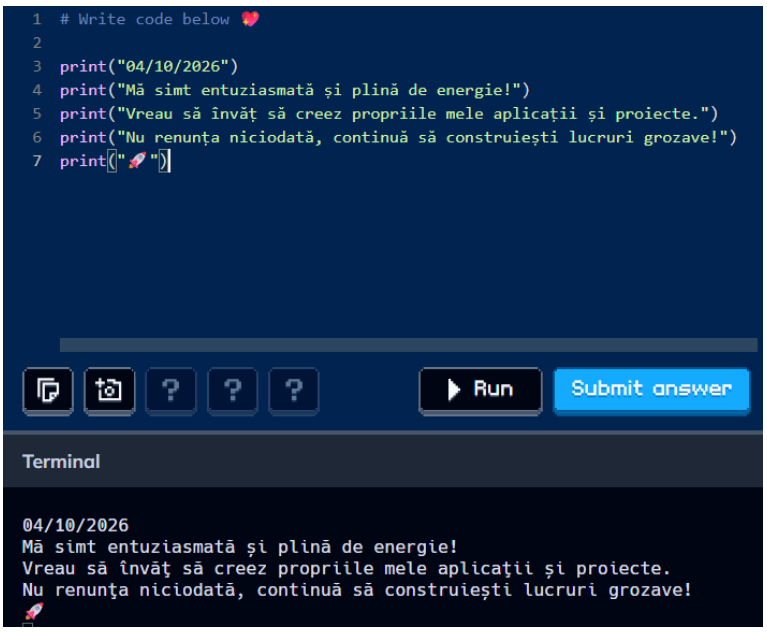


fig. 1.5 - Snail Mail

Capitolul 2 CODEDEX. Variables

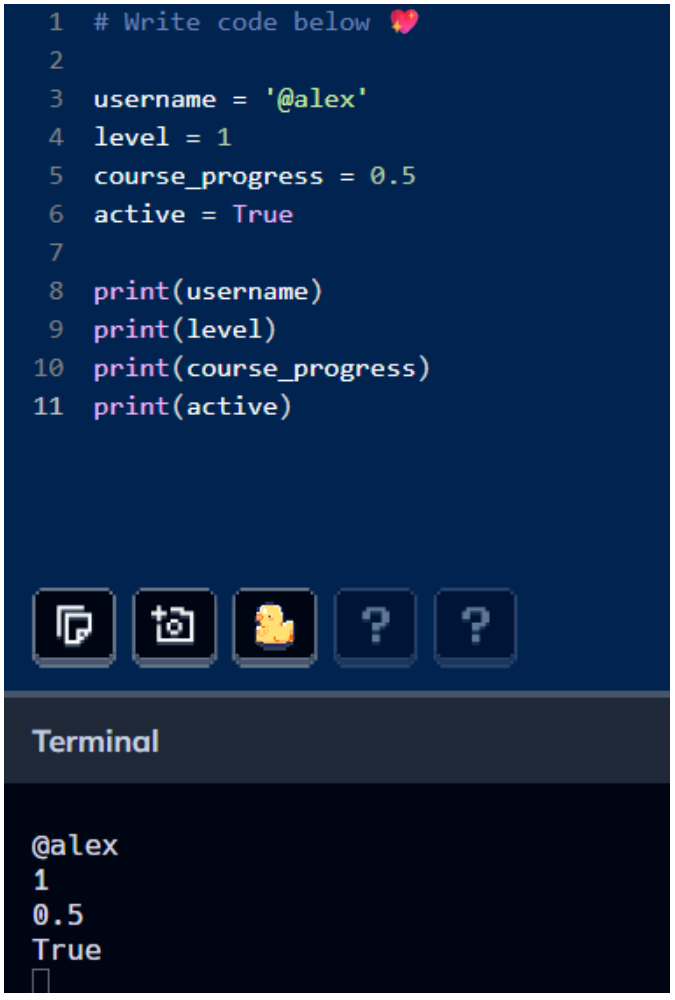


fig. 2.1 - Data Types

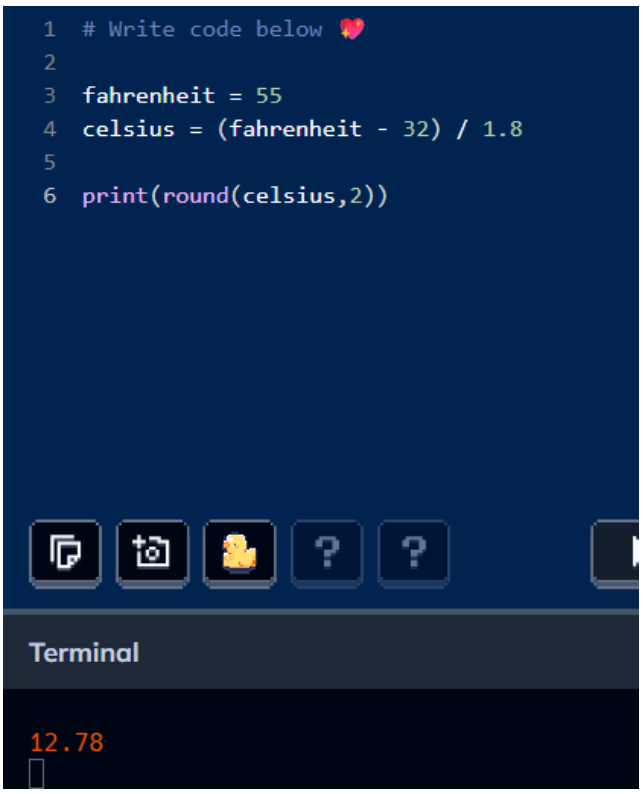


fig. 2.2 - Temperature

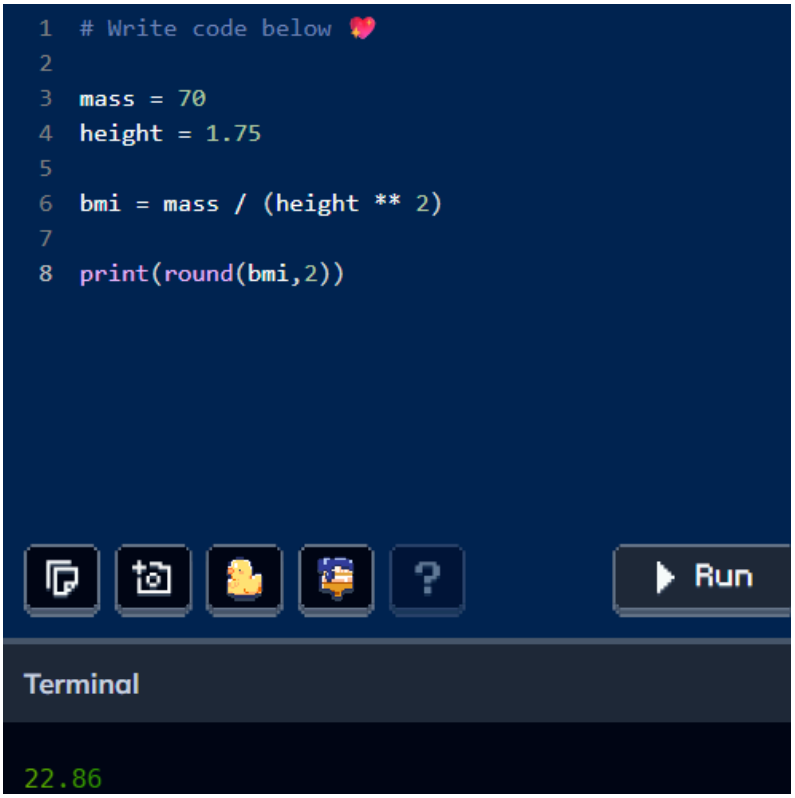


fig. 2.3 - BMI



fig. 2.4 - Pythagorean



fig. 2.5 Currency

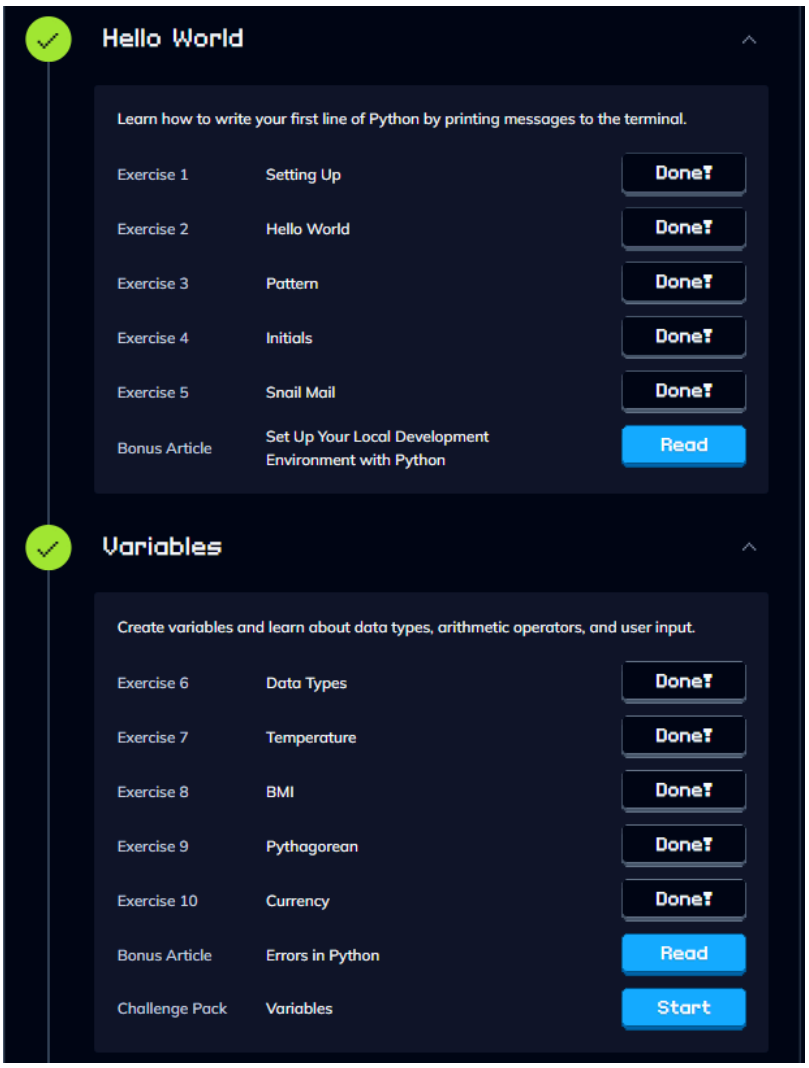


fig. 2.6 - Rezolvarea tuturor exercițiilor

Capitolul 3 LEARN PYTHON. Learn the basics.

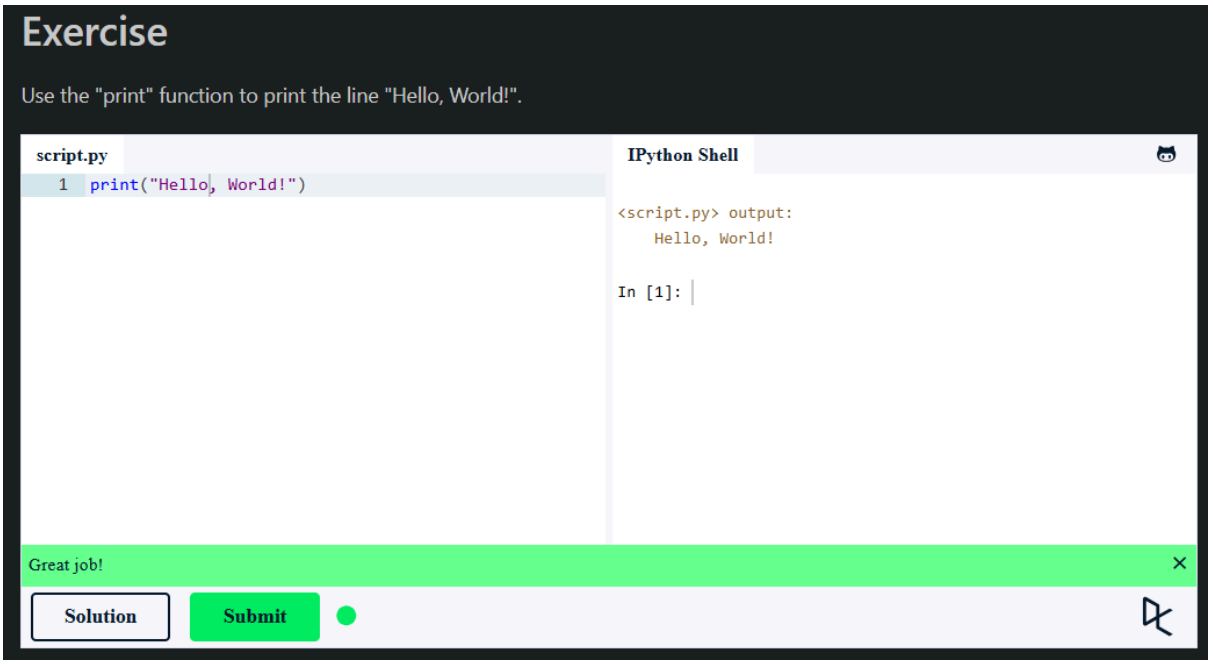


fig. 3.1 - Hello, Word



fig. 3.2 - Variables and Types

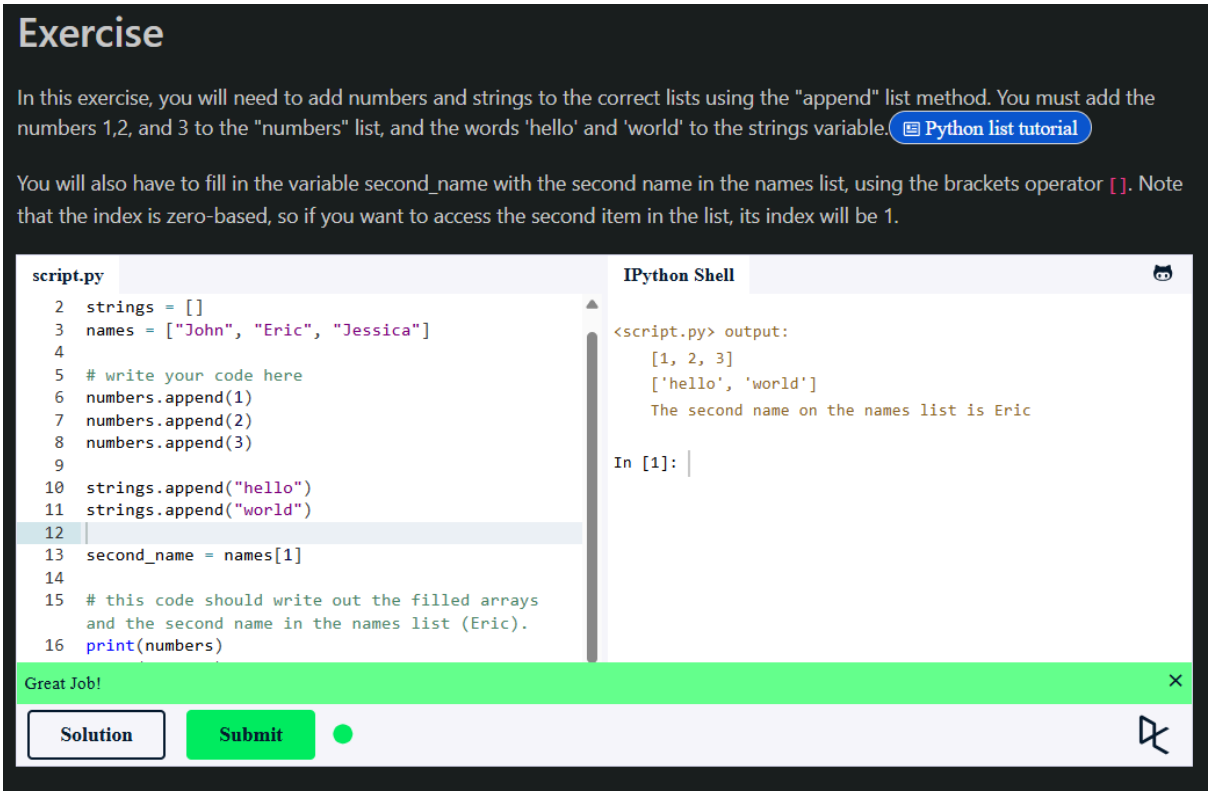


fig. 3.3 - Lists

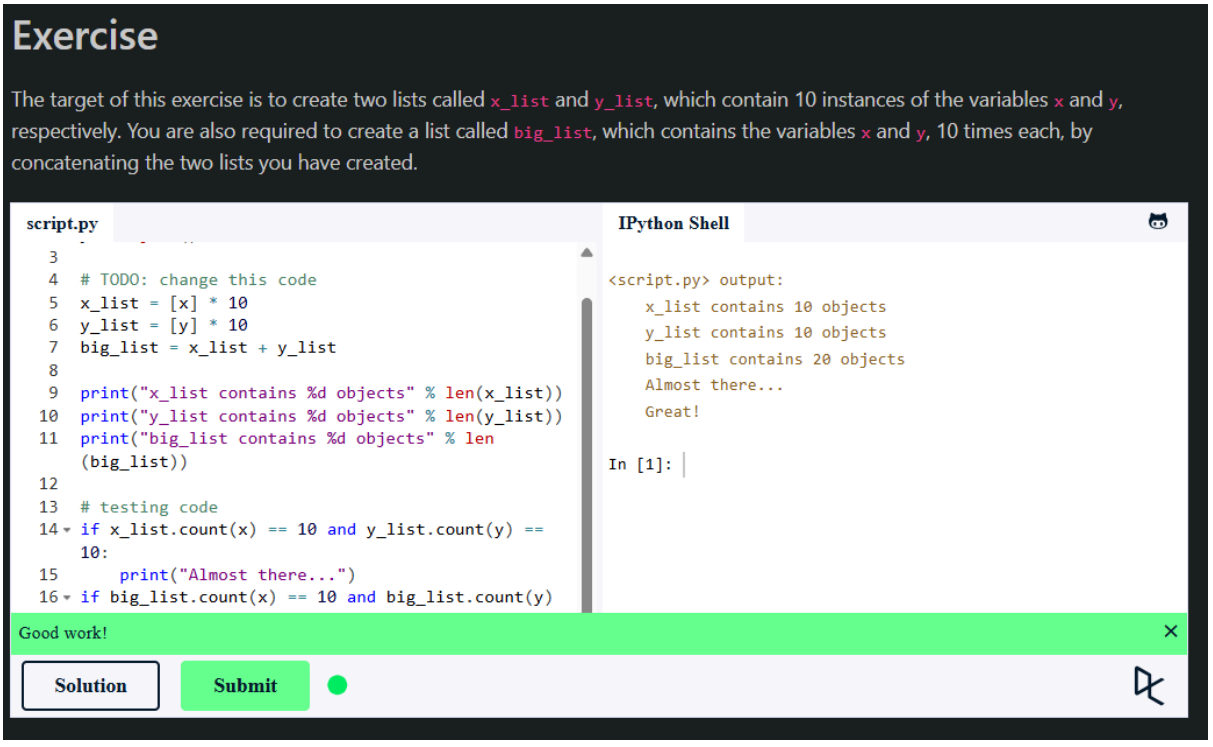


fig. 3.4 - Basic Operators

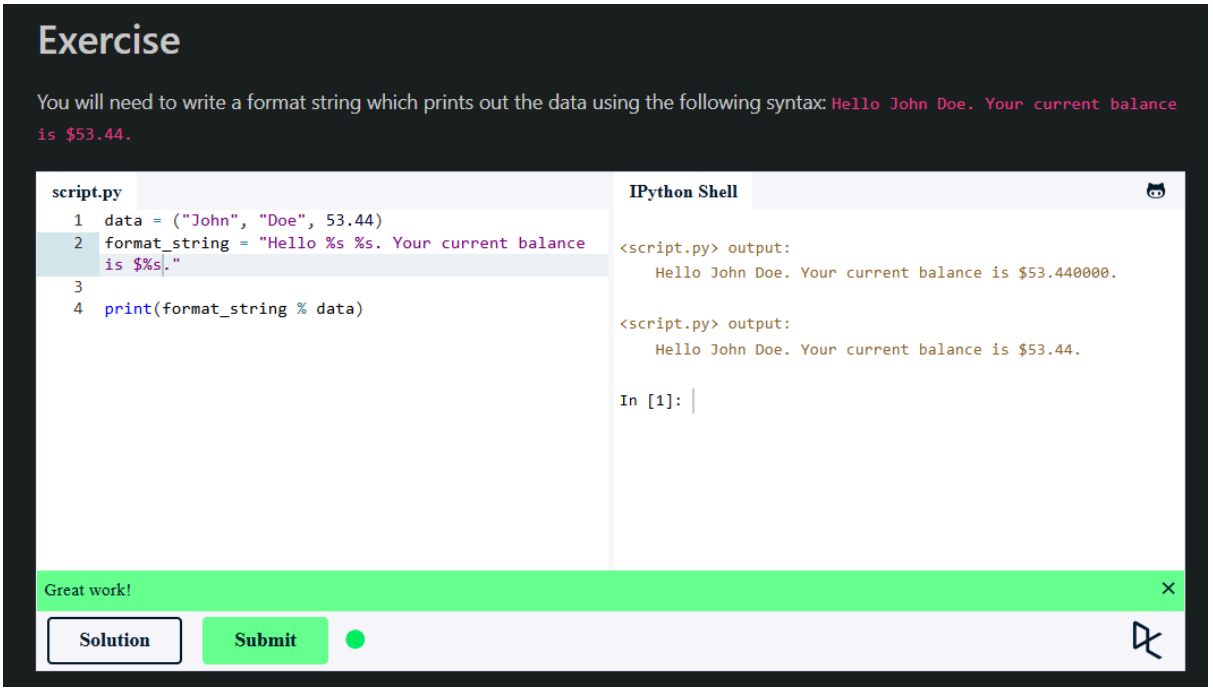


fig. 3.5 - String Formatting

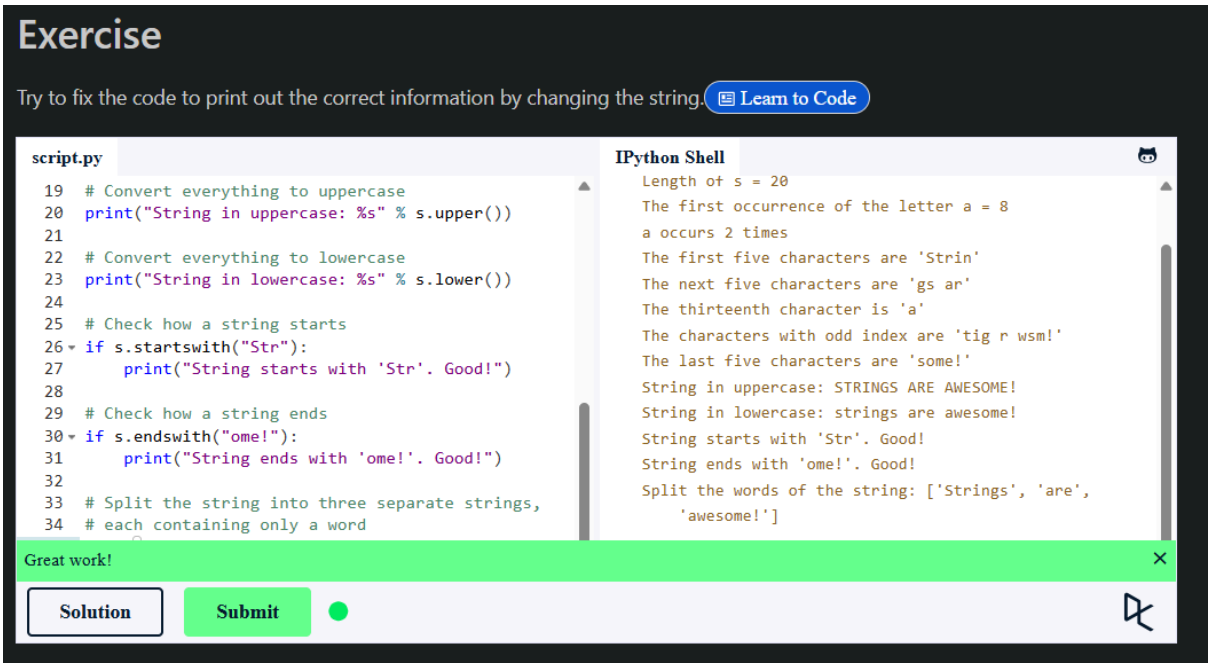


fig. 3.6 - Basic String Operations

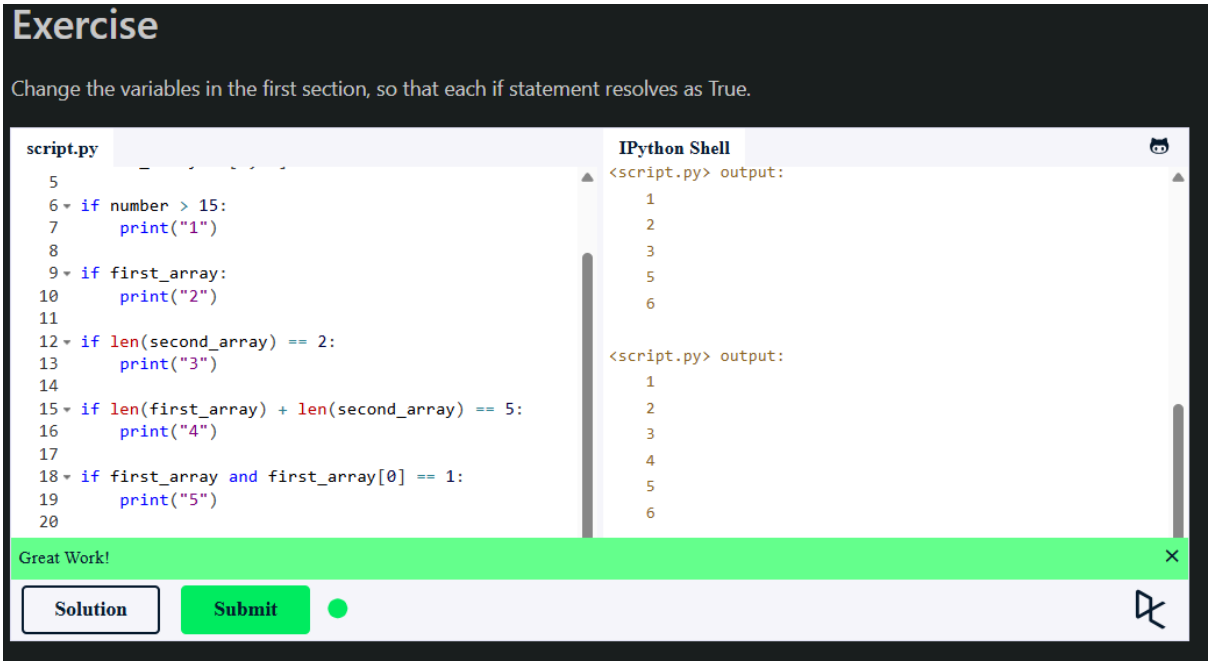


fig. 3.7 - Conditions

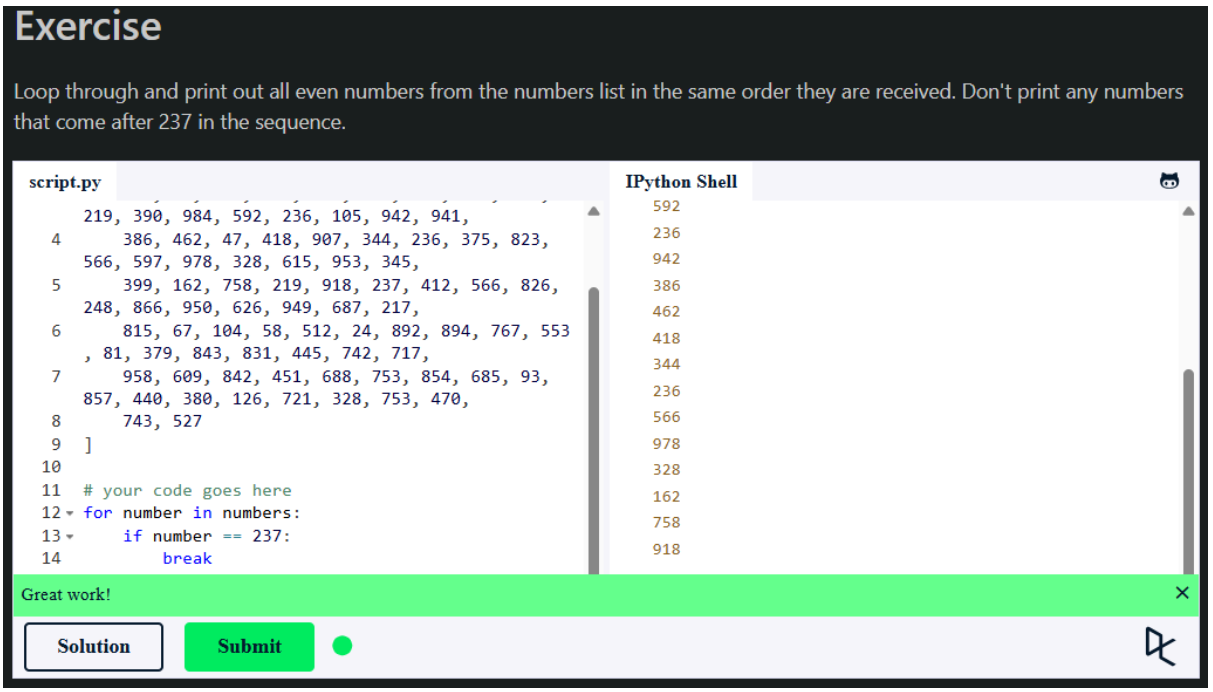


fig. 3.8 - Loops

Exercise

In this exercise you'll use an existing function, and while adding your own to create a fully functional program.

1. Add a function named `list_benefits()` that returns the following list of strings: "More organized code", "More readable code", "Easier code reuse", "Allowing programmers to share and connect code together"
2. Add a function named `build_sentence(info)` which receives a single argument containing a string and returns a sentence starting with the given string and ending with the string " is a benefit of functions!"
3. Run and see all the functions work together!

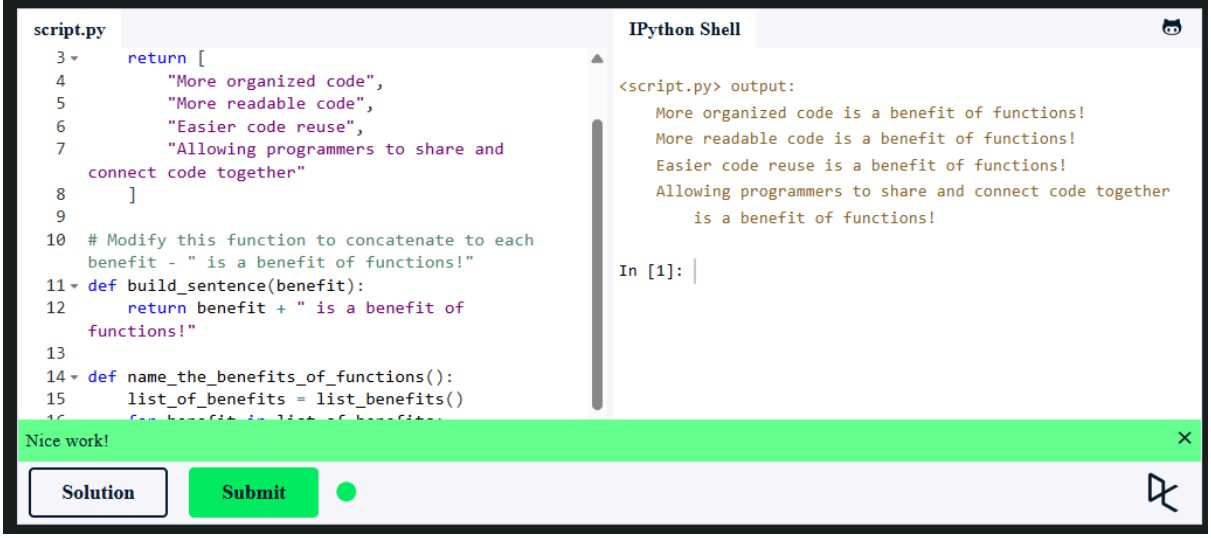


fig. 3.9 - Functions

Exercise

We have a class defined for vehicles. Create two new vehicles called car1 and car2. Set car1 to be a red convertible worth $60,000.00 with a name of Fer, and car2 to be a blue van named Jump worth $10,000.00.

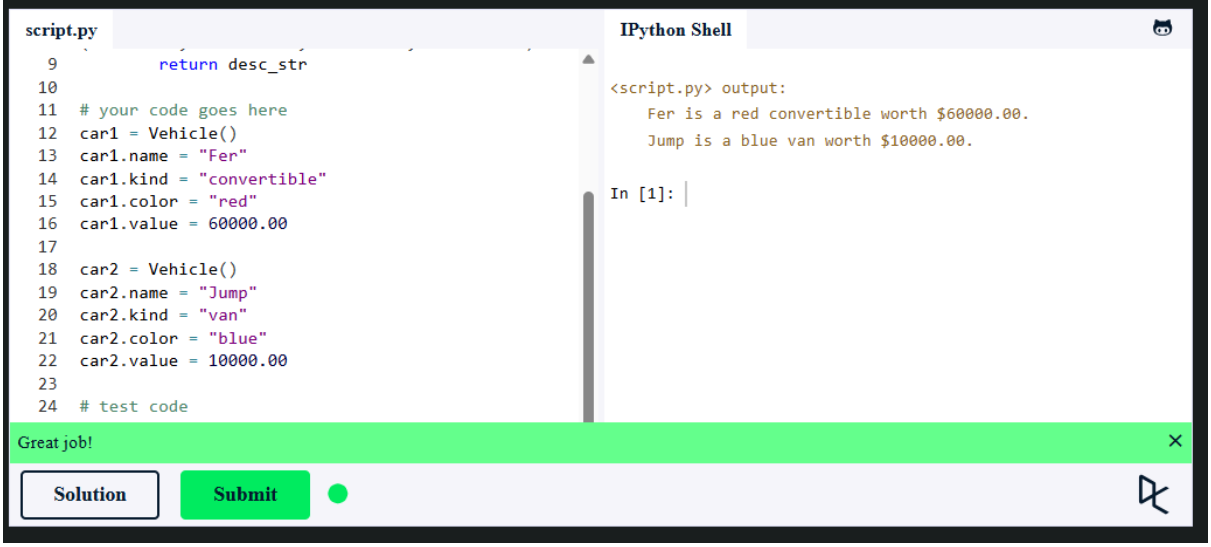


fig. 3.10 - Classes and Objects

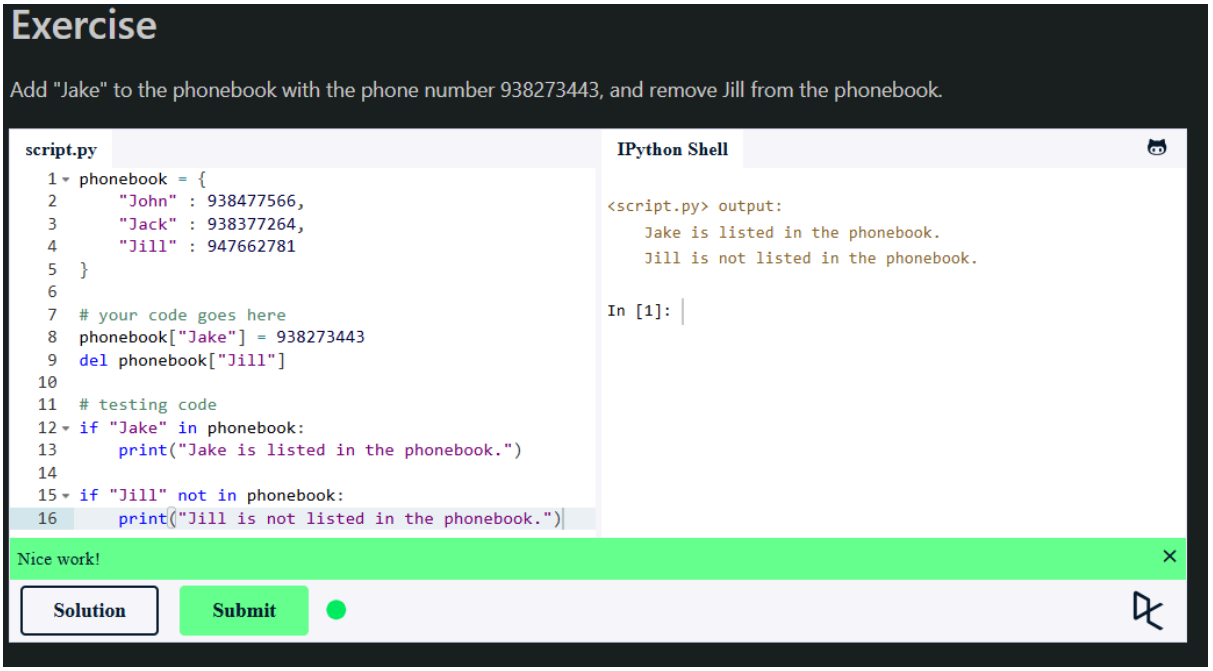


fig. 3.11 - Dictionaries

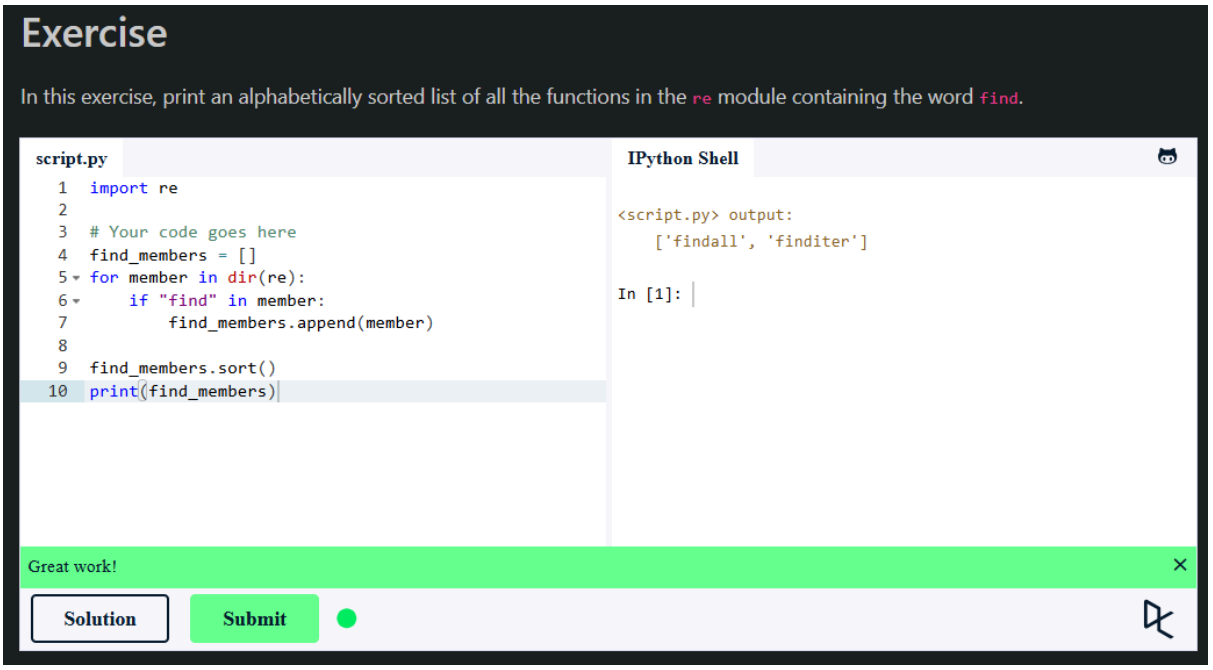


fig. 3.12 - Modules and Packages

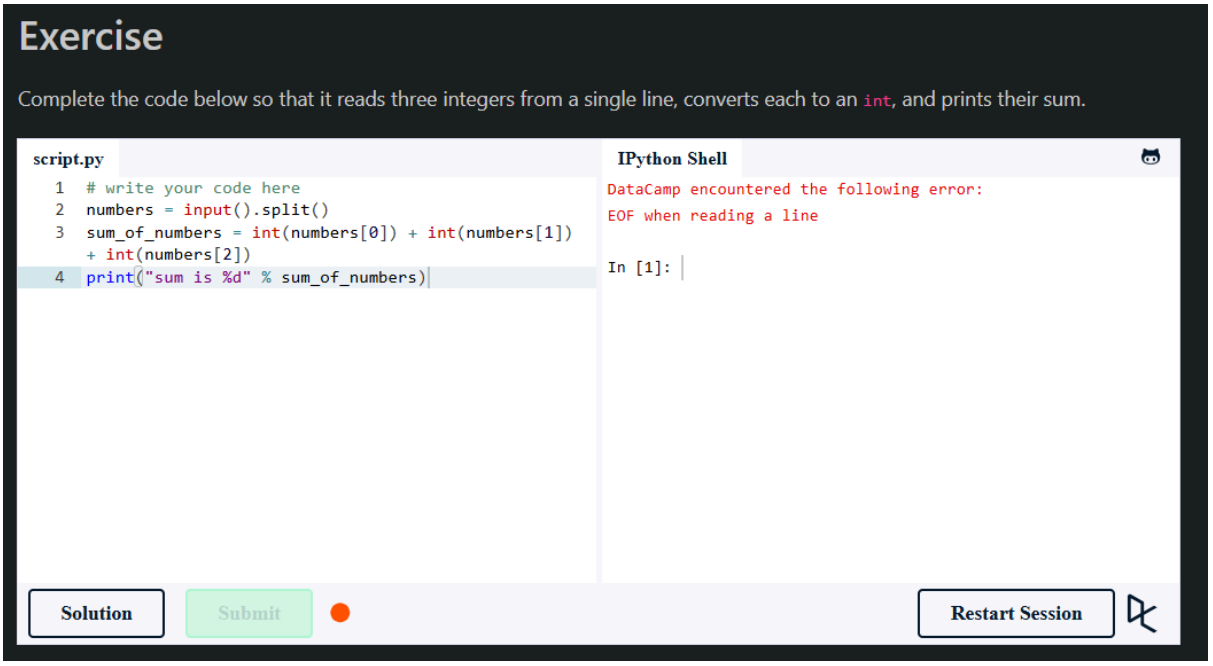


fig. 3.13 - Input and Output

Capitolul 4 LEARN PYTHON. Coding for kids



fig. 4.1 - Stating out



fig. 4.2 - Movement with functions



fig. 4.3 - Collecting items



fig. 4.4 - Pushing Objects



fig. 4.5 - Printing on screen



fig. 4.6 - Building objects

Capitolul 5 LEARN PYTHON. Advanced Tutorials

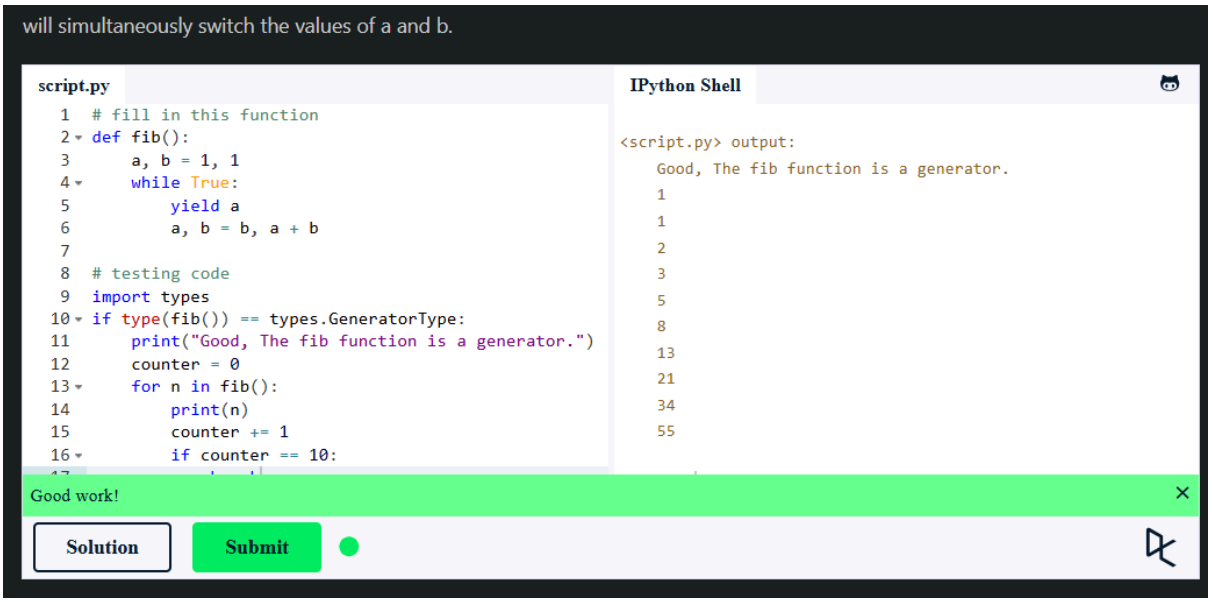


fig. 5.1 - Generators

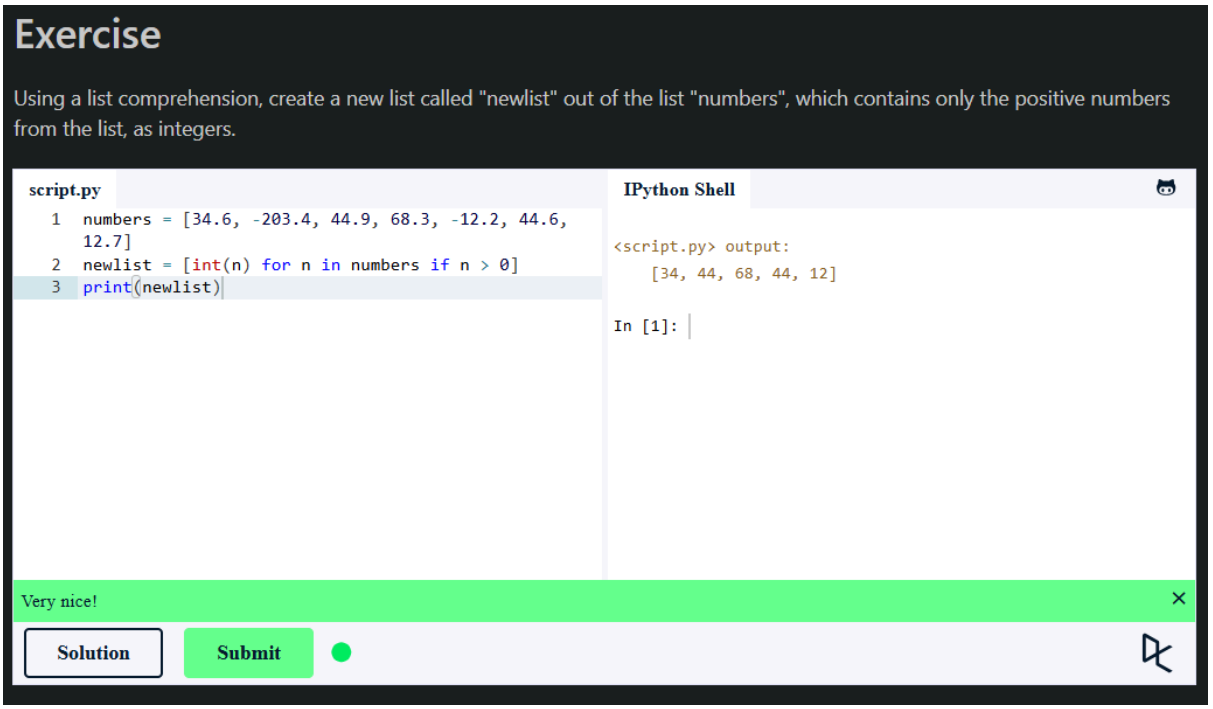


fig. 5.2 - List Comprehensions



fig. 5.3 - Lambda Functions

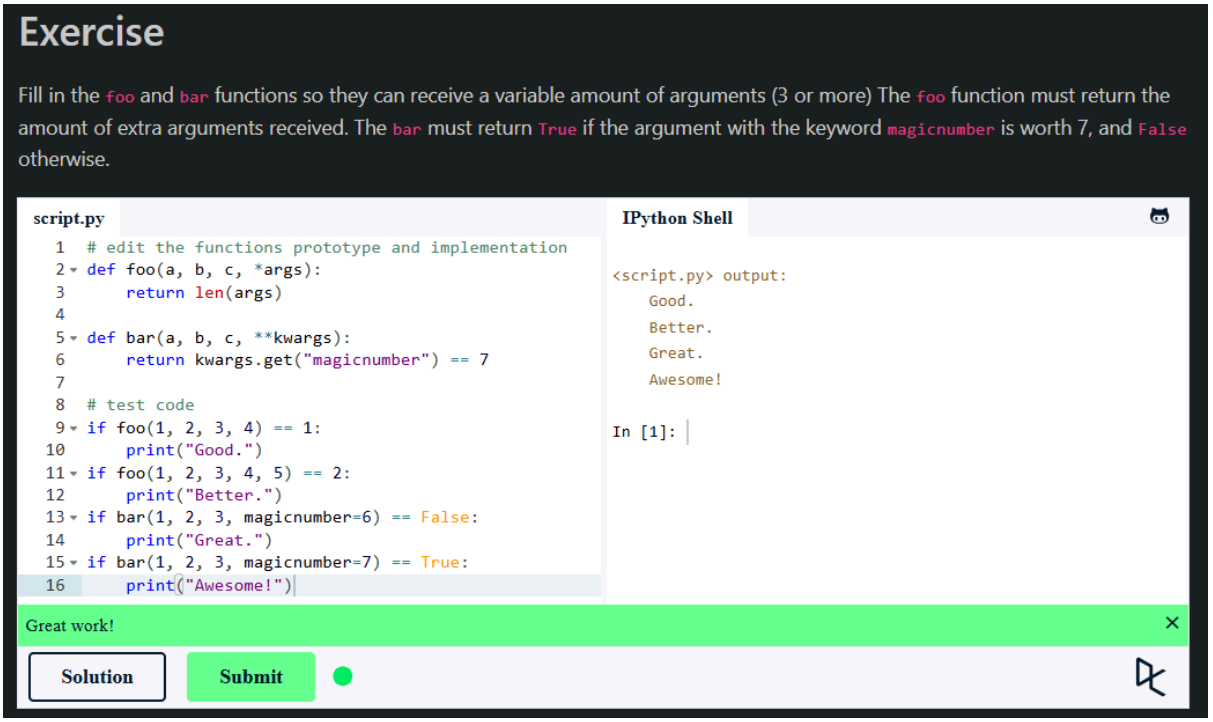


fig. 5.4 - Multiple Function Arguments

An example regex is `r"^(From|To|Cc).*?python-list@python.org"` Now for an explanation: the caret `^` matches text at the beginning of a line. The following group, the part with `(From|To|Cc)` means that the line has to start with one of the words that are separated by the pipe `|`. That is called the OR operator, and the regex will match if the line starts with any of the words in the group. The `.*?` means to un-greedily match any number of characters, except the newline `\n` character. The un-greedy part means to match as few repetitions as possible. The `.` character means any non-newline character, the `*` means to repeat 0 or more times, and the `?` character makes it un-greedy.

So, the following lines would be matched by that regex: `From: python-list@python.org To: !asp]<,. python-list@python.org`

A complete reference for the re syntax is available at the python docs.

As an example of a "proper" email-matching regex (like the one in the exercise), see this

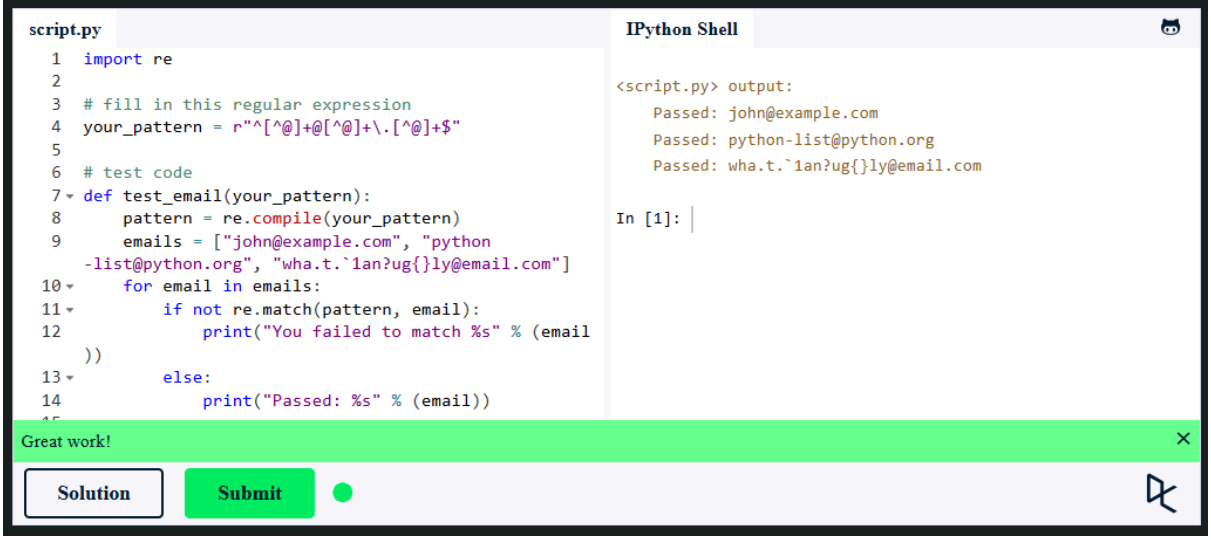


fig. 5.5 -Regular Expressions

Exercise

Handle all the exception! Think back to the previous lessons to return the last name of the actor.



fig. 5.6 - Exception Handling

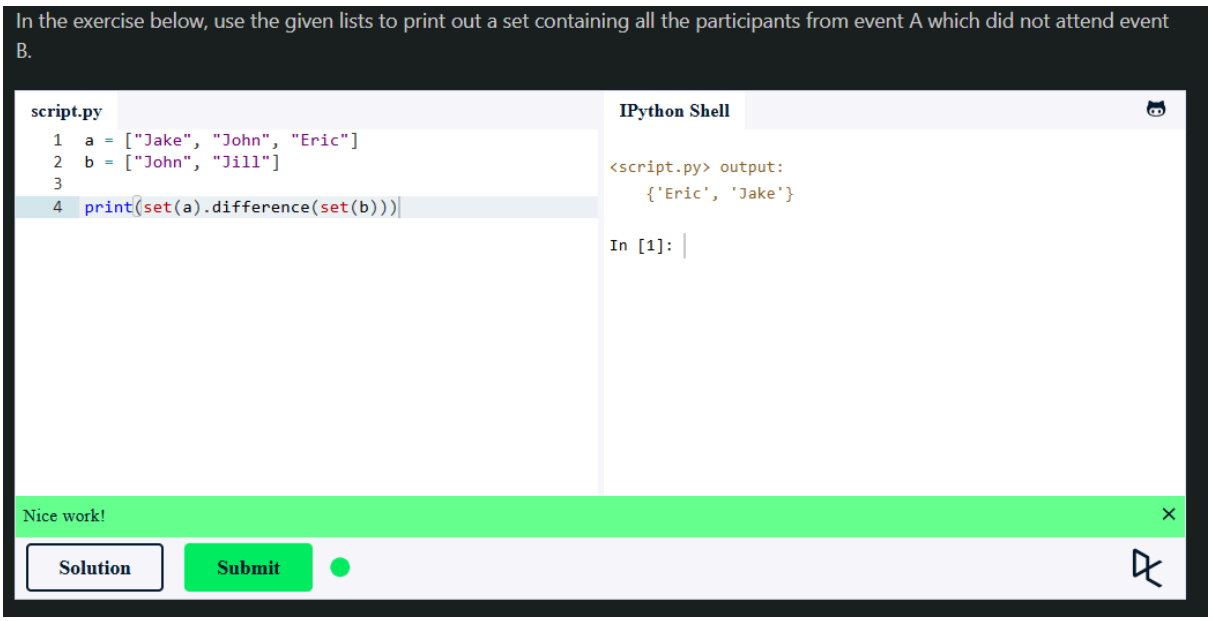


fig. 5.7 - Sets

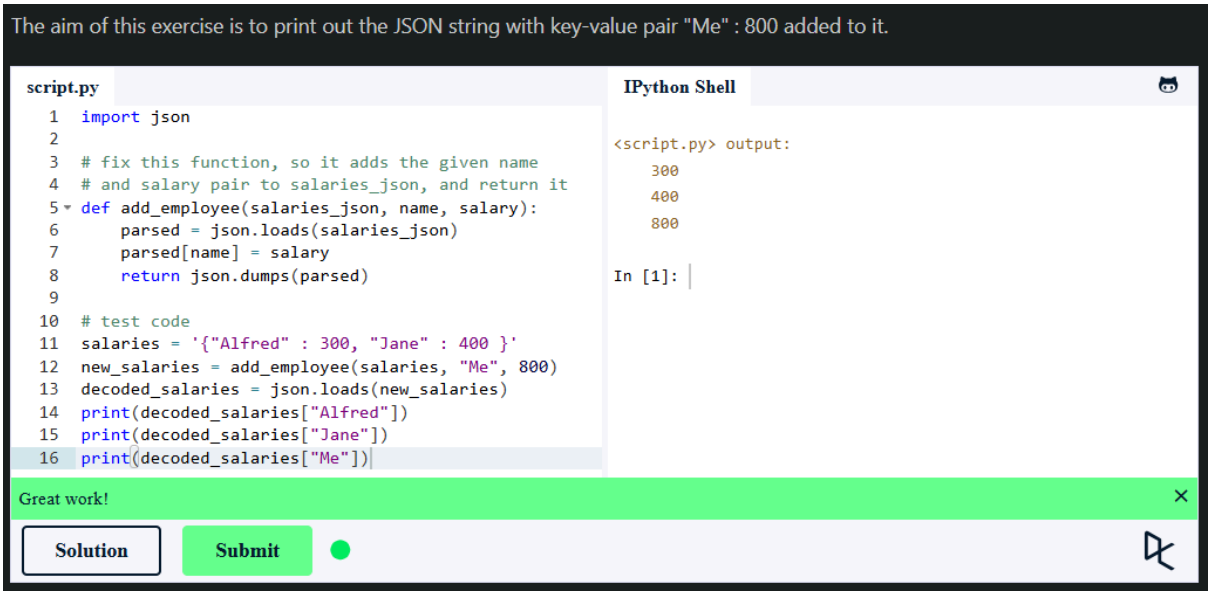


fig. 5.8 - Serialization

Concluzie

Privind în urmă la tot acest parcurs, de la primele linii de cod scrise în Python și gestionarea excepțiilor până la ghidarea personajului prin labirinturile acestor nivele interactive, simt că am făcut o trecere esențială de la simpla înțelegere teoretică la o gândire cu adevărat algoritmică. Fiecare obstacol întâmpinat fie că a fost vorba de o eroare de sintaxă, un pas greșit pe hartă, o piatră de împins sau un pod de construit m-a învățat să fiu mai atent la detalii, să îmi testez logica pas cu pas și să nu renunț atunci când codul nu funcționa din prima. Am înțeles că programarea nu înseamnă doar memorare, ci mai ales răbdare, creativitate și capacitatea de a transforma o problemă complexă într-o secvență de pași simpli și clari. Această experiență mi-a dat încredere și m-a echipat cu bazele solide de care aveam nevoie pentru a continua să explorez lumea fascinantă a tehnologiei.